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Review of Related Literature — 15-Minute Presentation Script

Slide 1 — Review of Related Literature

Speaker: JM

Good day, everyone.

For our Review of Related Literature, we examined previous studies involving three metaheuristic optimization algorithms:

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Simulated Annealing, Tabu Search, and Particle Swarm Optimization.

Our review focuses on how these algorithms work, what previous researchers have found about their strengths and weaknesses, and how they have been applied to three major problem areas:

route optimization, job scheduling, and machine-learning feature selection.

Rather than simply asking which algorithm is “best,” the literature shows that performance depends heavily on the type of problem, how solutions are represented, how the algorithms are configured, and how researchers evaluate them.

So throughout this review, we will focus not only on reported results, but also on the conditions and limitations behind those results.

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Slide 3 — Metaheuristic Optimization

Speaker: JM

To begin, metaheuristics are general search strategies designed to find high-quality solutions when checking every possible solution becomes impractical.

A central idea found throughout the literature is the balance between **exploration and exploitation**.

Exploration means searching different or unfamiliar parts of the solution space.

Exploitation means focusing more closely on areas where good solutions have already been found.

If an algorithm exploits too much, it may become trapped in a **local optimum**—a solution that is better than its nearby alternatives, but not necessarily the best overall.

On the other hand, if it explores too much, it may spend too much time moving around without improving.

Another major point from the literature is that there is **no single optimization algorithm that performs best on every possible problem**.

An algorithm that performs very well in routing may not perform equally well in scheduling or feature selection.

This is one of the major reasons why comparing these three algorithms across different application areas is useful.

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Slide 4 — Simulated Annealing Literature

Speaker: JM

The foundational work on **Simulated Annealing**, or SA, comes from Kirkpatrick, Gelatt, and Vecchi.

SA was inspired by the physical process of slowly cooling heated material.

In optimization, the algorithm begins with one solution and repeatedly creates a nearby alternative.

If the new solution is better, it is normally accepted.

However, what makes Simulated Annealing different from ordinary hill climbing is that it may also accept a **worse solution**.

Early in the search, when the temperature is high, worse solutions are accepted more easily. This allows the algorithm to escape local optima.

As the temperature decreases, it becomes increasingly selective and focuses more on refinement.

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The literature consistently identifies this as one of SA's main strengths: it is relatively simple, flexible, and can be adapted to many different problems.

However, several studies also identify **parameter sensitivity** as a major weakness.

Its performance can depend heavily on the starting temperature, cooling schedule, number of moves performed, stopping condition, and even the numerical scale of the objective function.

For example, Youssef and colleagues found that Simulated Annealing initially performed poorly in a VLSI floorplanning comparison. But part of the problem was that the objective values and temperature scale were poorly matched. After adjusting the scale, SA's performance improved substantially.

This is important because it shows that poor algorithm performance may sometimes result from poor calibration rather than from the algorithm itself.

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Slide 6 — Tabu Search Literature

Speaker: JM

Tabu Search, or TS, was formally developed by Fred Glover.

Like Simulated Annealing, it usually follows one current solution at a time.

However, instead of relying mainly on probability, Tabu Search uses **memory**.

Recent moves or characteristics of those moves are placed on a tabu list. This temporarily prevents the algorithm from simply undoing what it just did and repeatedly cycling through the same solutions.

It can also use an **aspiration criterion**, which allows a tabu move if that move produces an exceptionally good solution.

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The literature shows that one of the main strengths of Tabu Search is its use of memory to guide the search and avoid repeatedly returning to the same solutions.

Short-term memory helps prevent cycling by restricting recently used moves. Intermediate-term memory supports intensification, allowing the algorithm to focus more strongly on features that appear in high-quality solutions. Long-term memory supports diversification, encouraging the search to explore less-visited regions.

Because of these mechanisms, Tabu Search can perform strong local search while still moving beyond local optima.

However, this memory-based design is also one of its main weaknesses, because it introduces several important implementation decisions.

Researchers must determine the tabu tenure, neighborhood structure, candidate-list size, aspiration rule, and stopping condition. Watson and colleagues, for example, showed that tabu tenure affects how widely the algorithm explores. A tenure that is too short may allow cycling, while one that is too long may unnecessarily block useful moves.

So overall, Tabu Search's main strength is its ability to use search history to guide exploration and prevent cycling, while its main weakness is that its performance depends heavily on how its memory and neighborhood mechanisms are designed involving many parameters.

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Slide 8 — Particle Swarm Optimization Literature

Speaker: JM

The third algorithm is **Particle Swarm Optimization**, or PSO, originally introduced by Kennedy and Eberhart.

Unlike SA and TS, PSO is **population-based**.

Instead of following only one solution, it maintains a group of candidate solutions called particles.

Each particle remembers its own best experience and is also influenced by the best solution found by the group.

This allows the swarm to search several areas at the same time.

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One of the main strengths of PSO is its fast convergence and population-based exploration.

Because multiple particles search simultaneously and share information, the swarm can often move quickly toward promising solutions.

Later research also expanded PSO through inertia weights, different neighborhood structures, adaptive parameters, discrete versions, and hybrid methods.

However, one of its main weaknesses is premature convergence.

If too many particles follow the same promising solution too early, the swarm can lose diversity and become trapped around a local optimum.

Another weakness is that standard PSO was originally designed for continuous numerical spaces.

Problems such as route optimization, scheduling, and feature selection are usually discrete, so researchers often need modified versions such as binary PSO, geometric PSO, or other discrete representations.

So overall, PSO's main strengths are fast convergence and broad population-based search, while its main weaknesses are premature convergence and the need for special adaptations when applied to discrete problems.

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Slide 10 — Literature Review: Route Optimization

Speaker: Lance

The first practical application we reviewed is **route optimization**, especially the Traveling Salesman Problem.

The goal is simple: find the shortest route that visits every required location and returns to the starting point.

The literature shows that Simulated Annealing, Tabu Search, and Particle Swarm Optimization can all be used for this problem, but they improve routes in different ways.

Simulated Annealing

Simulated Annealing improves one route at a time by making small changes.

These changes may include swapping two cities, moving one city somewhere else in the route, or reversing part of the route.

Zhan and colleagues tested a modified version called **List-Based Simulated Annealing** on problems ranging from smaller routes to cases with tens of thousands of cities.

They found that the kind of route change used had a major effect on performance.

For example, **inversion**, which reverses the order of several cities in one section of the route, generally worked better than simply swapping two cities.

Using several kinds of route changes together produced the strongest overall results.

Tabu Search in Route Optimization

Speaker: Lance

Tabu Search also improves one route at a time, but its main advantage is that it **remembers recent route changes**.

For example, after changing a connection between two cities, it can temporarily prevent itself from immediately changing that connection back.

This helps the algorithm avoid repeatedly moving between the same routes and encourages it to continue searching for a better one.

Pirim and colleagues reviewed several routing studies and found that Tabu Search often produced strong route quality, especially as the number of locations increased.

However, Tabu Search may take more computation because it usually checks several possible route changes before deciding which one to use.

This means its memory can guide the search effectively, but evaluating many alternatives may make each step more expensive.

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Slide 11 — Particle Swarm Optimization in Route Optimization

Speaker: Lance

Particle Swarm Optimization approaches the problem differently because it searches using **several possible routes at the same time**.

Each candidate route can learn from routes that have already produced good results.

However, standard PSO was originally designed for ordinary numerical values, while a route depends on the exact order of its cities.

Researchers therefore use modified or discrete versions of PSO that can rearrange cities while keeping the route valid.

PSO is also often combined with a simple route-improvement method called **2-opt**.

In 2-opt, two connections in a route are removed and reconnected in a different way. This can remove unnecessary crossings and shorten the route.

The main advantage of PSO is that it can explore several different routes at once.

Its main limitation is that it needs more adaptation than SA or TS before it can properly work with ordered routes.

Route Optimization Synthesis

Overall, the reviewed literature suggests that each algorithm offers a different advantage.

Simulated Annealing is flexible and can improve routes using several simple types of changes.

Tabu Search uses memory to avoid repeating recent decisions and often performs strongly on larger route problems.

Particle Swarm Optimization explores several possible routes at the same time, but it requires a modified version or an added route-improvement method.

So the literature does not identify one method that is always best.

Their performance still depends on the size of the problem, the way routes are changed, the algorithm settings, and the amount of computation available.

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Slide 12 — Literature on Job Scheduling

Speaker: Lance

The second application is **job scheduling**.

Scheduling problems can involve deciding which machine handles a job, what order jobs should follow, and how to minimize outcomes such as makespan or tardiness.

The literature contains several different scheduling formulations.

Alharkan and colleagues compared Tabu Search, Geometric PSO, Simulated Annealing, Genetic Algorithm, and Iterated Local Search on a two-machine scheduling problem with a shared setup server.

Their results showed that Tabu Search performed particularly strongly on larger instances and most frequently reached the theoretical lower bound, while Geometric PSO generally ranked second.

Simulated Annealing and Genetic Algorithm were more competitive on smaller problems but their performance declined as problem size increased.

Other research has used modified discrete PSO for flexible job-shop scheduling, where an operation may be assigned to different eligible machines.

These studies show that PSO can work effectively in scheduling, but normally requires a representation designed specifically for discrete assignments and sequences.

Overall, Tabu Search appears frequently in scheduling literature because job swaps and sequence changes create natural neighborhoods that its memory mechanism can exploit.

However, the literature does not support saying that tabu search is always the best scheduling method, because results still depend on the scheduling formulation and implementation.

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Slide 13 — Literature on Machine-Learning Feature Selection

Speaker: Lance

The third major application is **machine-learning feature selection**.

Feature selection aims to remove irrelevant or redundant variables while keeping enough useful information for accurate prediction.

This creates a difficult search problem because a dataset with many features can have an enormous number of possible subsets.

The literature contains applications of all three algorithms.

Simulated Annealing can add or remove features from one candidate subset.

Tabu Search can prevent repeated addition and removal of the same variables.

Binary forms of PSO can represent each particle as a pattern of selected and unselected features.

Xie and colleagues proposed two enhanced PSO methods called PSOVA1 and PSOVA2.

They tested these methods on 13 datasets containing from only a few dozen features to more than 22,000.

Their results showed strong predictive performance, particularly for PSOVA2.

However, the methods did not always produce the smallest possible feature subsets.

This is an important lesson: reducing features is useful, but selecting fewer features is not automatically better if predictive performance becomes worse.

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Slide 14 — Allvi et al. and Methodological Weaknesses

Speaker: Julian

One especially useful study for our review is the work of Allvi and colleagues, who applied Simulated Annealing to feature selection for learning-to-rank.

The study demonstrates that SA can be adapted to search for useful feature subsets while evaluating ranking quality.

However, it also contains several important methodological limitations.

The paper does not clearly report complete Simulated Annealing settings, such as the exact cooling process and neighborhood configuration.

It also lacks strong repeated-run statistics.

The most serious concern is that candidate solutions appear to have been repeatedly evaluated using **test data during the optimization process**.

Normally, the test set should remain untouched until the final evaluation.

If an optimizer repeatedly uses test performance when selecting features, the chosen subset may gradually become tailored to that particular test set.

This can make reported performance appear better than it would be on truly unseen data.

We therefore do not treat this paper as strong evidence that SA is superior.

Instead, it is useful in two ways: it shows that SA can be applied to feature selection, and it highlights an experimental weakness that future studies should avoid.

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Slide 15 — Direct Comparisons Across Algorithms

Speaker: Julian

We also reviewed studies that directly compared two or more of the selected algorithms.

One example is Zhang and Nicholson, who compared two versions of Simulated Annealing, Tabu Search, and Particle Swarm Optimization for a network-flow parameter problem.

Interestingly, the algorithms produced similar final solution quality.

The main differences appeared in efficiency.

PSO generally improved more quickly and performed particularly well on smaller and larger test instances, while the Simulated Annealing variants required more computation.

Another useful comparison is the work of Youssef and colleagues.

Under a shared budget of 5,000 solution evaluations, Tabu Search performed best in their VLSI floorplanning experiments, followed by Genetic Algorithm and then Simulated Annealing.

However, as discussed earlier, Simulated Annealing was initially disadvantaged by poor scaling between its temperature and objective values.

Once this was corrected, its performance improved.

These studies reinforce an important lesson: **final solution quality alone does not tell the whole story.**

Two algorithms may reach similar solutions but require very different amounts of computation.

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Slide 16 — Common Limitations in Existing Literature

Speaker: Julian

Across the reviewed studies, several recurring methodological problems appear.

First is **parameter sensitivity.**

SA depends on cooling settings.

TS depends on memory and neighborhood settings.

PSO depends on population size and movement parameters.

If one algorithm receives careful tuning while another receives weak default settings, the comparison may be unfair.

Second is inconsistent computational conditions.

Different studies use different computers, programming languages, stopping rules, datasets, and evaluation budgets.

Third, some papers report only one run or only the best result.

This is problematic for stochastic algorithms because one lucky run may not represent normal performance.

Fourth, some studies use weak baselines.

A new algorithm may appear strong simply because it was compared only with random selection or a very simple method.

Finally, hybrid algorithms create another difficulty.

If PSO is combined with SA, Tabu Search, local search, crossover, and several adaptive mechanisms, stronger performance does not tell us which component caused the improvement.

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Slide 17 — Synthesis of the Literature

Speaker: JM

When we combine the findings of the reviewed literature, several broad patterns emerge.

Simulated Annealing is relatively simple and flexible. Its ability to accept temporary worsening solutions helps it escape local optima. However, it is especially sensitive to its temperature schedule, objective scaling, and neighborhood design.

Tabu Search makes stronger use of local structure and search history. It frequently performs well in routing and scheduling problems where good neighboring moves can be clearly defined. However, it requires careful decisions about memory and restrictions.

Particle Swarm Optimization provides population-level exploration and often converges quickly. However, it can lose diversity prematurely, and discrete problems require modifications to the standard PSO mechanism.

Most importantly, the literature does **not** show one universal winner.

Performance changes according to problem type, problem size, encoding, parameter settings, computational budget, stopping conditions, and implementation quality.

So rather than asking simply, “Which algorithm is best?”, the stronger question is:

Which algorithm appears most appropriate under particular problem conditions?

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Slide 18 — Research Gap and Direction of the Study

Speaker: JM

This leads to the research gap identified from our review.

There is already a large amount of research on these algorithms.

The problem is that the findings are scattered across different problem domains and different experimental conditions.

One paper may evaluate route distance.

Another focuses on runtime.

Another uses objective evaluations.

Another uses a modified or hybrid version of the algorithm.

Some provide repeated-run statistics, while others report only one outcome.

Because of these differences, it is difficult to combine previous findings into clear practical guidance.

Our study therefore aims to provide a **structured comparative review** of Simulated Annealing, Tabu Search, and Particle Swarm Optimization across route optimization, job scheduling, and machine-learning feature selection.

We will compare the patterns reported by previous researchers, while also considering the methodological strengths and weaknesses behind those results.

We also plan to conduct small sample tests as supplementary demonstrations.

These tests are not intended to prove a universal winner.

Instead, they will help us observe whether the behavior we see in simplified examples supports or differs from the patterns reported in the literature.

Ultimately, our goal is to develop a comparative framework that explains the strengths, weaknesses, and practical suitability of each algorithm.

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So that is all for our literature review Thank you for listening

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Here are our references

Week 11

1 Oco

2 Orientation 2

3 lexical grammar

4 specs of the school thing

5 people show their school specs

6 stalgcm review 1

7 stalgcm review again

9. He sick (absent)

10. He sick (half day)

11. Genetic shit full day

12. Presentation of paper

13. Presentation of lit rev